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GE Medical Systems

Technical Publications

Direction 2108265
Revision 2

Signa[®] Torso Array Coil

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/ Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

<FTP://3.87.40.2/GLOBEPRO/QUALSYS/DOCS/190016MF.PDF>

Rev. 11/15/2000

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- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

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- ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
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- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
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- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
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- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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1- INTRODUCTION

1-1 How the Torso Array Coil Operates

This coil is electrically and mechanically a combination of the Pelvic Array and the General Purpose Flex coil. It will replace the Body Coil for imaging the abdomen (including the liver, pancreas and kidney) and the thoracic region (pulmonary and cardiac). See Illustration 1-1.

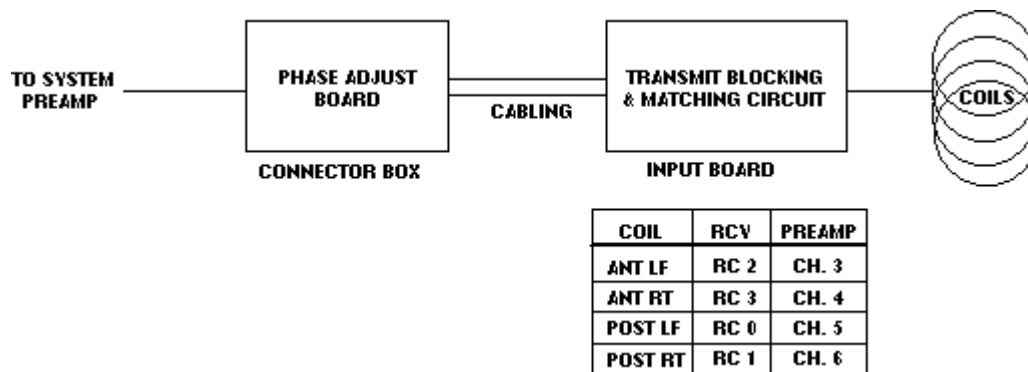


ILLUSTRATION 1-1
TORSO ARRAY FUNCTIONAL BLOCK DIAGRAM

The Torso multicoil is a four coil array to obtain the higher SNR of smaller MRI surface coils and retain a large field of view. Also, because the array is split in two halves (one in the front and one in the back) there is a "volume" affect that increases the SNR at the center of the body. Each half consists of two individual coils that are approximately 8 inches by 12 inches with a 1.4 inch overlap. This translates to a coil area of approximately 14 inches (35.6 cm R/L) by 12 inches (30.5 cm S/I) overall for each half. The array is intended to be used in one mode only utilizing all four coils.

The interface to Signa is a 30 pin Bendix connector with a double banana plug mechanical enhancement. This connection plugs into the multicoil port and is keyed for a specific field strength use only. The two halves are connected to each other via cabling. A means to reduce cable loops will be provided with Velcro cable ties which will allow adjustment for individual patients.

The array is designed for feet first abdominal imaging. This implies a cable length of approximately 6 feet to fit a patient with a 36 inch inseam.

The only active components, including the PIN diodes, are on the input boards via an access cover over the input circuitry of the coil. The design will not allow for PIN diode replacement in the field at this time.

Each coil element must be turned off during transmit. This is accomplished with active blocking which includes a tuned L-C network and a single PIN diode per coil element. The input capacitor is chosen such that the transmit blocking impedance (at 250 mA DC bias to the diode) is greater than 5k ohm. This meets the design requirements of 3.4 ohm/cm*cm. The coil element size is 20.32 cm (8 in) X 30.48 cm (12 in) which is 619.35 cm*cm. Therefore the minimum blocking impedance necessary is 2.1k ohm.

The 5k ohm allows a two times safety factor.

Since the input capacitor is chosen for optimum blocking protection, a matching circuit is necessary for a loaded 50 ohm match. The matching circuit consists of a shunt capacitor and series inductor. The matching circuit will be fixed components with no tuning necessary. The blocking circuitry and matching circuit reside on the M/C Torso Input Bd.

A 50 ohm phase shift network ensures a multiple of a half wavelength from the 30 pin connector to the coil input. This phase compensation network takes into account cable lengths and the matching circuit. The phase shift networks for all four coils reside on the Torso Phase Adjust Board located in the Quick Disconnect Enclosure.

1-2 Compatibility

The Torso Array Coil is compatible with the following hardware configurations:

- Signa® Advantage™ 1.5T, 1.0T, & 0.5T System (1.5T only)
- Signa® Advantage™ System (1.5T only)

1-3 Related Documents

- Direction 15400, Signa® Advantage™ 1.5T, 1.0T, & 0.5T System
- Direction 15401, Signa® Advantage™ 1.5T, 1.0T, & 0.5T Subsystems
- Direction 15404, Signa® Advantage™ 1.5T, 1.0T, & 0.5T Renewal Parts
- Direction 15300, Signa® Advantage™ System

1-4 Organization of this Document

This manual is divided into the following sections:

Section 1, Introduction, describes how the Torso Array Coil operates and when and where the Torso Array Coil can be used.

Section 2, Setup and Calibration, describes installation procedures.

Section 3, Functional Checks, describes the normal power-up sequence.

Section 4, Replacement / Maintenance, describes field maintenance procedures.

Section 5, Renewal Parts, lists field replaceable parts.

1-5 Environmental Requirements

Operate and store the Torso Array Coil in the Scanner Room.

2- SETUP AND CALIBRATION

2-1 Checking the Shipping List

Table 2-1 lists the M1087TA Signa 1.5T Torso Array Coil parts. Check that all parts have been shipped.

TABLE 2-1
1.5T TORSO ARRAY COIL SHIPPING LIST

<u>QTY.</u>	<u>ITEM</u>	<u>PART NUMBER</u>
1	1.5T Torso Array Flex Coil	2104700
1	Torso Replaceable Cover Set	2106940
1	Service Manual	2108265

2-2 Installing the Torso Array Coil

At the Boot Terminal, install a new soft key. This key will be used by the operator to select TORSO (elements 1-4) imaging. Name the key "TORSO". See Illustration 2-1.

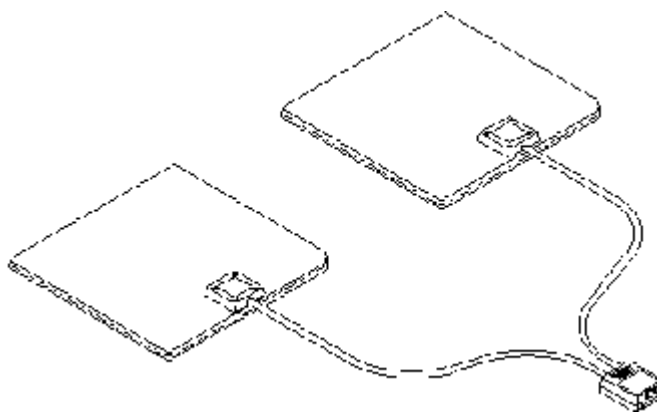


ILLUSTRATION 2-1
TORSO ARRAY COIL

Refer to the Signa & Signa Advantage System Manuals for information on installing soft keys (use Coil/Phased Array default values in Table 2-2 and 2-3).

TABLE 2-2
COIL VALUES

<u>SYSTEM</u>	<u>COIL NAME</u>	<u>COIL TYPE</u>	<u>EXTREMITY COIL</u>	<u>CABLE LOSS</u>	<u>COIL LOSS</u>	<u>RECON SCALE FACTOR</u>	<u>XMIT COIL</u>	<u>MULTI-COIL</u>
1.5T	TORSO	SURF	NO	1.05	0.313	Head Coil Recon Scale Factor x 1.00	QUAD	YES

TABLE 2-3
PHASED ARRAY VALUES

<u>MULTICOIL NAME</u>	<u>NUMBER OF RECEIVERS</u>	<u>START RECEIVER</u>	<u>STOP RECEIVER</u>	<u>PORT ENABLE MASK</u>	<u>ERROR ENABLE MASK</u>
TORSO	4	0	3	6	6

2-3 Functional Checks

1. Perform a Body coil scan SNR verification. Refer to Section 3-1, BODY COIL SNR VERIFICATION.
2. Perform a Torso Array Coil SNR Verification. Refer to Section 3-2, TORSO ARRAY COIL SNR VERIFICATION.

2-4 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks outlined below to ensure that the coil is operating properly:

1. Check the external cable for cracks or breaks once a week. Refer to Section 4-5, CHECKING THE CABLES.
2. Perform a coil SNR verification. Refer to Section 3-2, TORSO ARRAY COIL SNR VERIFICATION.
3. Record the date and value calculated in Section 3-3, SNR IMAGE ANALYSIS in column 2 under "SNR Data QA Check" of the Data Table as is instructed.
4. As is instructed in the Data Table, divide the SNR value obtained in the periodic QA check by the original SNR value and record in column 6 of the Data Table.
5. If this ratio is not greater than 85%, then there may be a problem in the coil system. Contact your local GE Service Representative.

3- FUNCTIONAL CHECKS

3-1 Body Coil SNR Verification

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to *System Level Procedures: Troubleshooting: TLT Procedure*.

Phantom Required

- Body TLT Phantom, 46-265635G6 or Body SPT Phantom, 2298119
- Body TLT Loader, 46-287902G1 or SPT Short Loader, 2125244

Setup Procedure

Caution

The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

1. Remove Quad Head Coil (if present) from cradle.
2. Select **[NEW STUDY] (4.x)** or **[New Exam] (5.x)** to allow a new Landmark to be set.
3. Position the Body Phantom in the center of the Body Loader at the center of the cradle. Landmark the center of the phantom and advance to isocenter using the **[ADVANCE TO SCAN]** button. See Illustration 3-1.

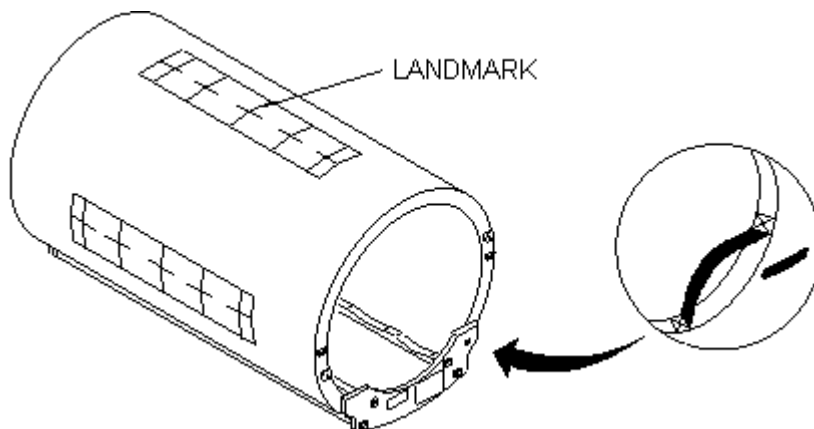


ILLUSTRATION 3-1
BODY PHANTOM/LOADER LANDMARK SETUP

4a. **For 4.x:** Setup Scan Prescription as shown in Table 3-1.

TABLE 3-1
BODY COIL SNR - SCAN PROTOCOL (4.X)

Id:	geservice	Echo Time (TE):	[20 msec]
Name:	body snr	Rep Time (TR):	[2000 msec]
Patient Weight:	300	Auto CF:	[Peak]
Patient Entry:	[Head First]	Field of View:	[48 cm]
Patient Position:	[Supine]	Scan Thickness:	[3 mm]
Axial/Sag. Landmark:	[Sternal Notch]	Scan Location:	(S/I) 0
Coil Type:	[Body Coil]	FOV Center:	(R/L) R0 (A/P) A0
Scan Plane:	[Axial]	Acq. Matrix:	[256 x 256]
Image Mode:	[Single Scan]	Frequency Direction:	[A/P]
Pulse Sequence:	[Multiple Echo]	Imaging Time:	[1 NEX 8:58]
Imaging Options:	[None]	Contrast:	[No]
or enter PSD		Table Delta:	0mm
Filename			
Number of Echoes:	[1]		

4b. **For 5.x:** Setup Scan Prescription as shown in Table 3-2.

TABLE 3-2
BODY COIL SNR - SCAN PROTOCOL (5.X)

Id:	geservice	Rep Time (TR):	[2000 msec]
Name:	body snr	Auto CF:	[Peak]
Patient Weight:	300	Field of View:	[48 cm]
Patient Entry:	[Head First]	Scan Thickness:	[3 mm]
Patient Position:	[Supine]	Interscan Spacing:	[Other] 0
Axial/Sag. Landmark:	[Sternal Notch]	Start Loc (I/S):	0 End Loc (I/S): 0
Coil Type:	[Body Coil]	No. of Scan Location:	1
Scan Plane:	[Axial]	FOV Center	(L/R):0 (A/P) 0
Image Mode:	[2D]	Acq. Matrix (freq.):	[256]
Pulse Sequence:	[Spin Echo]	Acq. Matrix (phase)	[256]
Imaging Options:	[None]	Frequency Direction:	[A/P]
or enter PSD Filename		Imaging Time:	[1 NEX 8:58]
Number of Echoes:	[1]	Contrast:	[No]
Echo Time (TE):	[20 msec]	Table Delta:	0 mm

5. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.
6. Select **[Scan]**. Observe the resulting images. Ensure that there are no artifacts of any sort in the resulting image. Record the Exam number and Series number for SNR Calculations.
7. Select **[Scan]** again. This second image will be used for determination of Body Coil mode SNR.

8. Select **[Cancel]**. Refer to Section 3-3 for SNR image analysis.

3-2 Torso Array Coil SNR Verification

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to *System Level Procedures: Troubleshooting: TLT Procedure*.

Phantom Required

- Head TLT Phantom, 46-265826G6
- Head TLT Loader, 46-287899G1

Setup Procedure

1. Select **[NEW STUDY] (4.x)** or **[New Exam] (5.x)** to allow a new Landmark to be set.

Caution

The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

2. Remove Quad Head Coil (if present) from cradle.
3. Place Torso Array Coil to be tested around the body loader/body phantom with the Torso coils place on the left and right side of loader. See Illustration 3-2.
4. Attach the velcro straps around the external cables to secure the two cables close to each other.
5. Connect Torso Array Coil connector to its mating connector in the Carriage Assembly.

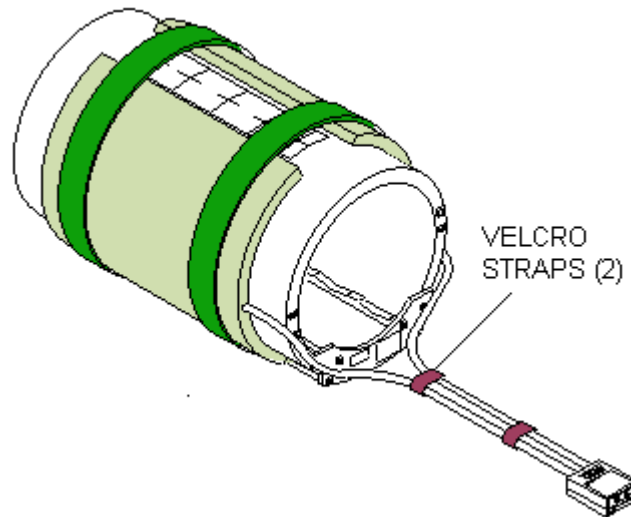


ILLUSTRATION 3-2
TORSO ARRAY/BODY LOADER SETUP

6. Position the Torso Array/loader in the center of the cradle. Landmark for center of body phantom and advance to isocenter using the **[ADVANCE TO SCAN]** button. See Illustration 3-3.

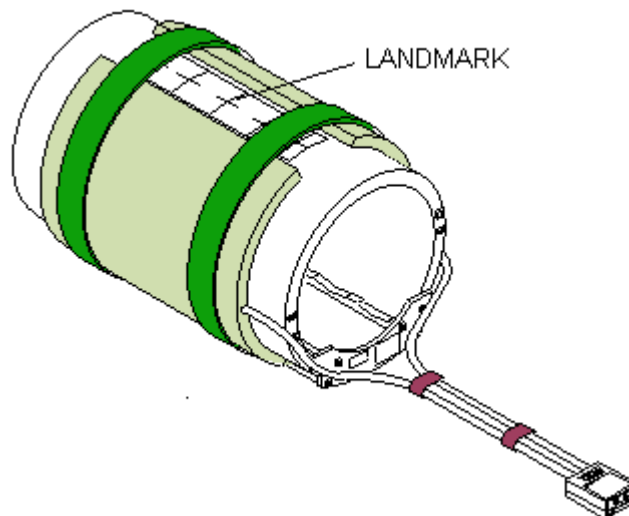


ILLUSTRATION 3-3
TORSO ARRAY/BODY LOADER LANDMARK

Phased Array Coil Receiver Paths

The Phased Array Coils have multiple individual antenna loop elements inside the assembly; the TPS chassis has multiple receivers (RCV0-RCV3). See Illustration 3-4.

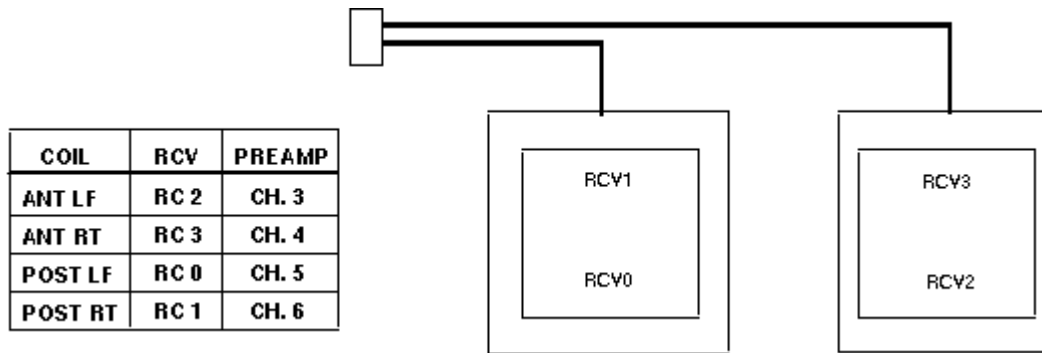


ILLUSTRATION 3-4

TORSO ARRAY COIL TO TPS RECEIVER CORRESPONDENCE

7a. **For 4.x:** Setup Scan Prescription as shown in Table 3-3.

TABLE 3-1
BODY COIL SNR - SCAN PROTOCOL (4.X)

Id:	geservice	Echo Time (TE):	[20 msec]
Name:	torso snr	Rep Time (TR):	[2000 msec]
Patient Weight:	300	Auto CF:	[Peak]
Patient Entry:	[Head First]	Field of View:	[48 cm]
Patient Position:	[Supine]	Scan Thickness:	[3 mm]
Axial/Sag. Landmark:	[Sternal Notch]	Scan Location:	(S/I) 0
Coil Type:	[Other Coils] [Torso]	FOV Center:	(R/L) R0 (A/P) A0
Scan Plane:	[Axial]	Acq. Matrix:	[256 x 256]
Image Mode:	[Single Scan]	Frequency Direction:	[A/P]
Pulse Sequence:	[Multiple Echo]	Imaging Time:	[1 NEX 8:58]
Imaging Options:	[None]	Contrast:	[No]
or enter PSD		Table Delta:	0mm
Filename			
Number of Echoes:	[1]		

7b. **For 5.x:** Setup Scan Prescription as shown in Table 3-4.

TABLE 3-2
BODY COIL SNR - SCAN PROTOCOL (5.X)

Id:	geservice	Rep Time (TR):	[2000 msec]
Name:	body snr	Auto CF:	[Peak]
Patient Weight:	300	Field of View:	[48 cm]
Patient Entry:	[Head First]	Scan Thickness:	[3 mm]
Patient Position:	[Supine]	Interscan Spacing:	[Other] 0
Axial/Sag. Landmark:	[Sternal Notch]	Start Loc (I/S):	0 End Loc (I/S): 0
Coil Type:	[Other Coils] [torso]	No. of Scan Location:	1
Scan Plane:	[Axial]	FOV Center	(L/R):0 (A/P) 0
Image Mode:	[2D]	Acq. Matrix (freq.):	[256]
Pulse Sequence:	[Spin Echo]	Acq. Matrix (phase)	[256]
Imaging Options:	[None]	Frequency Direction:	[A/P]
or enter PSD Filename		Imaging Time:	[1 NEX 8:58]
Number of Echoes:	[1]	Contrast:	[No]
Echo Time (TE):	[20 msec]	Table Delta:	0 mm

8. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.
9. Select **[Scan]**. Observe the resulting image of the sphere. See Illustration 3-5 (normal image). Ensure that there are no artifacts of any sort in the sphere image. Record the Exam number and Series number for SNR Calculations.
10. Select **[Scan]** again. This second image of the sphere will be used for determination of Torso mode SNR.
11. Select **[Cancel]**. Refer to Section 3-3 for SNR image analysis.

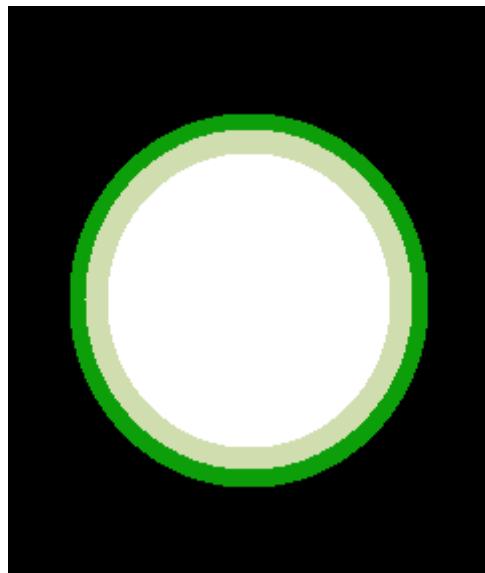


ILLUSTRATION 3-5
TORSO ARRAY COIL IMAGE

3-3 SNR Image Analysis

3-3-1 SNR Image Analysis (Release 4.x) (See Section 3-3-2 for Release 5.x)

Description

The CLIPS macro will ask for the first image in the series to analyze. The macro assumes you have done two back-to-back scans, and will take the designated image and the next image in the series (i.e. if you give it image 3, it will take 3 and 4). It will also ask you to designate three points on the border of the phantom image to use in positioning the analysis ROI. If your image is too big, too small, or the three points were selected too close together, the macro will inform you.

Procedure

1. Touch [UTILITIES], then [Clips].

CLIPS

```
-----
1 Run CLIPS
2 Save files
3 Restore files
4 Remove files
5 Exit
```

Enter the number of your selection:1 [ENTER]

```
1 = Operator console
2 = Remote1
3 = Remote2
4 = Exit
```

Which image processor would you like to use (1 2 or 3) ? ...1 [ENTER]

IP selected is at the operator console.

Do you wish to boot the Image Processor? (Y or N)Y [ENTER]
(You do not have to reboot the IP each time you enter.)

Welcome to the Command Line Image Processing System (CLIPS)

CLIPS >LIST(STUDY); [ENTER]

STUDY LIST

```
=====
STUDY #      PATIENT NAME      PATIENT I.D.      DATE
-----
00001        BDY_SNR            GESERVICE         9-JULY-89
00002        HD_SNR            GESERVICE         9-JULY-89
=====
```

CURRENT DEFAULTS --- PATIENT ID = / /

```
N - NEXT PAGE
P - PREVIOUS PAGE
S - SELECT STUDY
C - CANCEL
```

choice ?.....S [ENTER]

ENTER STUDY NUMBER --.....1 [ENTER]
(Enter appropriate number.)

CLIPS >LIST(SERIES); [ENTER]

SERIES LIST

```
=====
SERIES #      # OF IMAGES      SERIES TYPE      PULSE TYPE
=====
```

001	002	AXIAL	ME
002	002	SAG	ME

CURRENT DEFAULTS --- PATIENT ID = SNR 0000S/00S/001

N - NEXT PAGE

P - PREVIOUS PAGE

S - SELECT SERIES

C - CANCEL

choice ?S [ENTER]

ENTER SERIES NUMBER --.....1 [ENTER]
(Enter appropriate number.)

CLIPS >DIS; [ENTER]
(or you could LIST (IMAGE); to select the image you want.)

CLIPS >EXECUTE(SNR); [ENTER] or just EXE(SNR); [ENTER].

THIS MACRO CALCULATES THE SNR USING THE GE HEAD OR BODY PHANTOM.
ENTER THE NUMBER OF THE FIRST IMAGE OF THE TWO-IMAGE SET AND !

EXAMPLE INPUT> 1!

Please start entering data. Type an ! to end input mode.

CLIPS INPUT>1! [ENTER]

CORRECT POSITIONING OF THE SNR ANALYSIS ROI DEPENDS ON FINDING THE CENTER OF THE PHANTOM. YOU WILL BE PROMPTED TO IDENTIFY THREE POINTS AT THE EDGE OF THE PHANTOM. CARE SHOULD BE TAKEN TO PICK THREE POINTS AS FAR AWAY FROM EACH OTHER AS POSSIBLE (I.E. 2, 6, AND 10 O'CLOCK) AND AVOID FLAT SPOTS AND IRREGULARITIES ON THE EDGE.

See Illustration 3-6.

Note

The order of cursor placement must be 10, 6, then 2 o'clock positions. Using a different order may give an error.

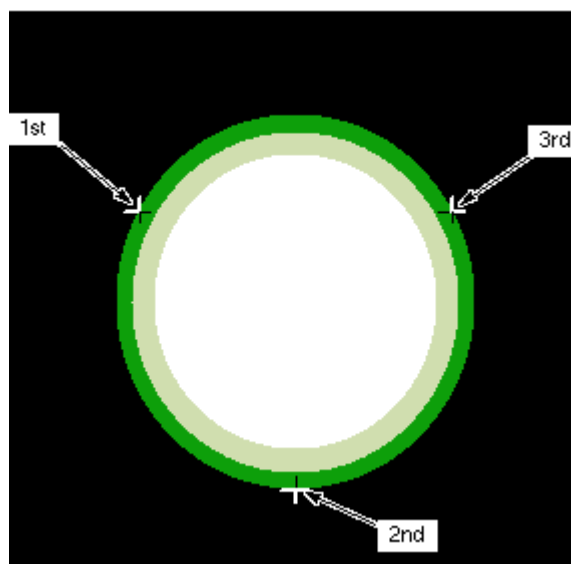


ILLUSTRATION 3-6
SNR CURSOR POSITIONS

POSITION CURSOR AT FIRST POINT AND ENTER ! (Place cursor at 10:00 position on edge of image.)

Please start entering data. Type an ! to end input mode.

CLIPS INPUT>! [ENTER]

POSITION CURSOR AT SECOND POINT AND ENTER ! Place cursor at 6:00 position.

Please start entering data. Type an ! to end input mode.

CLIPS INPUT>! [ENTER]

POSITION CURSOR AT THIRD POINT AND ENTER ! Place cursor at 2:00 position.

Please start entering data. Type an ! to end input mode.

CLIPS INPUT>! [ENTER]

PHANTOM POINTS AND RADIUS OK. BEGIN ANALYSIS.

The program will then perform the analysis and display the final values on the image monitor. Record Signal, Noise and Signal to Noise in Data Table.

DO YOU WANT TO SAVE THIS DATA?

ENTER A 1! FOR YES, A 0! FOR NO.

Please start entering data. Type an ! to end input mode.

CLIPS INPUT>..... 0! [ENTER]

CLIPS >LIST(SERIES); [ENTER]

Select the next series and repeat the above analysis for each image pair. Type BYE; when completed to exit CLIPS.

2. When all analyses are complete, reset Auto Center Frequency mode by touching **[SCAN MODES]**, **[Default Auto CF]**, then **[EXECUTE]**.

Note

You do not have to reset the Auto Center Frequency mode if you are proceeding on to more image quality scans.

3. Record the date and value calculated in the appropriate column under "SNR Data QA Check" of the Data Table (following Section 5, RENEWAL PARTS) as is instructed.

3-3-2 SNR Image Analysis (Release 5.x)

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first and the same ROI is used to calculate noise from the subtracted image. The signal value, noise value, and signal to noise ratio are reported. There is an option to save the difference image with the results annotated.

Procedure

1. Touch **[UTILITIES]**, **[MR Tools]**, **[Image Quality]**, then **[SNR Test]**. The SNR Test screen is displayed; see Illustration 3-7.
2. Enter image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

☐ Note: Accept changes to continue only after an analysis has been performed.

ILLUSTRATION 3-7
SNR TEST SCREEN

3. If high pass filtering is desired to be performed on data, touch **[Filter Off]** which will highlight and change to **[Filter On]**.
4. If the difference image annotated with data is to be created, touch **[Image Off]** which will highlight and change to **[Image On]**.
5. Touch **[Accept]** to begin analysis. The final values are displayed on the touch screen, see Illustration 3-7 (see previous screen).
6. Touch **[Continue]** then select the next exam and repeat the above analysis for each image pair.
7. Record the date and value calculated in the appropriate column under "SNR Data QA Check" of the Data Table (following Section 5, RENEWAL PARTS) as is instructed.

3-4 Checking the PIN Diodes with Digital Multimeter (DMM)

Note

There is one PIN diode for each antenna element. This procedure will indicate if the PIN diode is defective.

1. Select the DIODE TEST function on the Digital Multimeter (DMM).
2. Connect the POSITIVE lead of the DMM to the external cable connector row B, Pins 5, 7, 9, or 11. Connect the NEGATIVE lead to the external cable connector row A, Pin 1 (ground). Refer to Table 3-5.

TABLE 3-5
DIODE TEST CONNECTIONS

COIL NUMBER	POSITIVE LEAD CONNECTION		NEGATIVE LEAD CONNECTION	
	FOR STEP 2	FOR STEP 4	FOR STEP 2	FOR STEP 4
ANT - LEFT	B5	A1	A1	B5
ANT - RIGHT	B7	A1	A1	B7
POST - LEFT	B9	A1	A1	B9
POST - RIGHT	B11	A1	A1	B11

3. A reading of 0.400 to 0.600 should be observed on the DMM.
4. If a reading below 0.400 is observed in either direction, either the output cable is shorted or bad PIN diode.
5. If a reading above 0.600 is observed in step 2, the PIN diode is defective.
6. Connect the POSITIVE lead of the DMM to the external cable connector row A, Pin 1 (ground). Connect the NEGATIVE lead to the external cable connector row B, Pins 5, 7, 9, or 11. Refer to Table 3-5 (see previous screen).
7. A reading of INFINITY should be observed on the DMM.
8. If a reading of INFINITY is observed in both directions, either the output cable is open or the PIN diode is open.
9. If any of the above conditions fails, replace the coil. **The PIN diodes are not FRU's.**

3-5 Checking the External Cable

1. Select the DIODE TEST function on the Digital Multimeter (DMM).
2. Connect the POSITIVE lead of the DMM to the external cable connector row B, Pins 5, 7, 9, or 11. Connect the NEGATIVE lead to the external cable connector row A, Pin 1 (ground). Refer to Table 3-6.

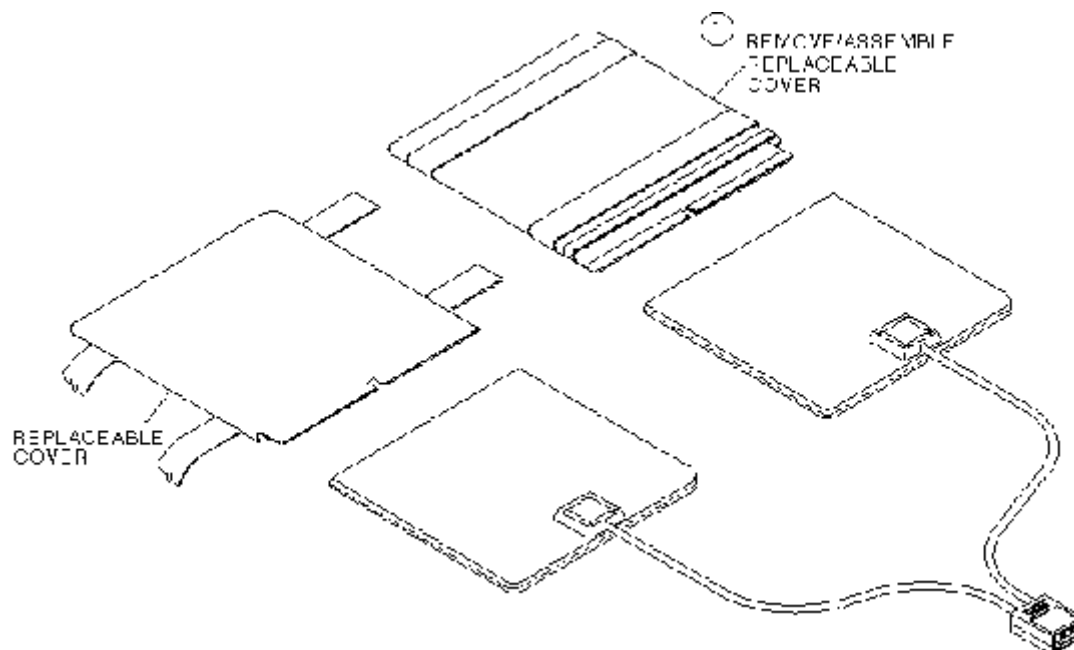
TABLE 3-6
EXTERNAL CABLE TEST CONNECTIONS

COIL NUMBER	POSITIVE LEAD CONNECTION		NEGATIVE LEAD CONNECTION	
	FOR STEP 2	FOR STEP 4	FOR STEP 2	FOR STEP 4
ANT - LEFT	B5	A1	A1	B5
ANT - RIGHT	B7	A1	A1	B7
POST - LEFT	B9	A1	A1	B9
POST - RIGHT	B11	A1	A1	B11

3. Flex the appropriate external cable, especially near the connectors and the strain relief, and observe that a reading of 0.400 to 0.600 should remain on the DMM, with no instability or fluctuations.
4. Connect the POSITIVE lead of the DMM to the external cable connector row A, Pin 1 (ground). Connect the NEGATIVE lead to the external cable connector row B, Pins 5, 7, 9, or 11. Refer to Table 3-6 (see previous screen).
5. Flex the appropriate external cable, especially near the connectors and the strain relief, and observe that a reading of INFINITY should remain on the DMM, with no instability or fluctuations.
6. If either cable fails any of the above tests, replace the coil. **The external cables are not FRU's.**

4- REPLACEMENT / MAINTENANCE

4-1 Disassembly / Assembly of Torso Array Coil



4-2 Replacing the External Cable

Note

The external cable can not be replaced in the field. Order a new Torso Array Coil.

4-3 Replacing the PIN Diodes

Note

The PIN diodes can not be replaced in the field. Order a new Torso Array Coil.

4-4 Replacing the Mechanical Hardware

1. Replace the entire Torso Array Coil. Refer to Section 5, Renewal Parts, for Torso Array Coil part number.

4-5 Checking the External Cable

Check the external cables for cracks or breaks once each week. Replace the entire Torso Array Coil if any damage or wear is found to the external cables.

4-6 Cleaning the Coil

Caution

Avoid damaging sensitive electronic parts. Do not spray or pour dishwashing solution directly onto the Torso Array Coil, or external cable. Do not use alcohol to clean the Torso Array Coil. Never submerge the Torso Array Coil in any liquid.

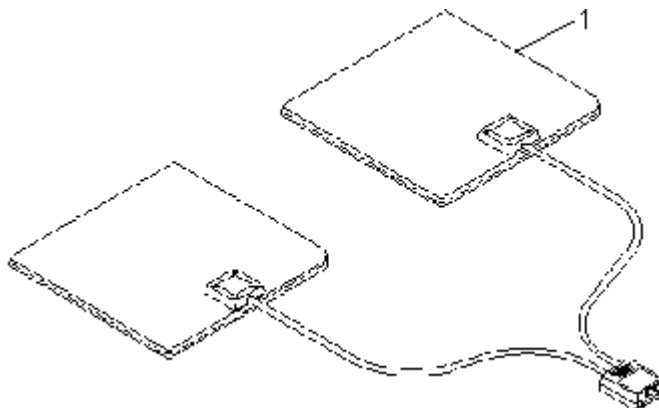
Clean the Torso Array Coil and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.

The replaceable covers can be cleaned with a 10% bleach solution.

5 - RENEWAL PARTS

1.5T Torso Array Coil

2104700



<u>Item</u>	<u>Part Number</u>	<u>FRU</u>	<u>Name</u>	<u>Qty.</u>	<u>Description (Remarks)</u>
1	2104700	1	TORSO COIL	1	1.5T TORSO ARRAY COIL

Use the space provided below to record the calculated signal to noise ratio (SNR) data obtained from Section 3, Functional Checks. After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2 - 4 below and the date they were obtained in column number 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Body Coil SNR Value: _____

Torso Coil SNR Value: Anterior_____Posterior_____Both_____

Date:

[illegible]

Make additional copies of this document as is needed.

DEFECTIVE COIL RETURN FORM

To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality related complaints with a description of scan protocol used.

Date _____
 Site Name _____
 Site Address _____
 Service Engineer _____
 Coil Serial Number _____
 Date Coil Installed _____
 Description of Coil Problem _____

ELECTRICAL CHECKS PIN Diode Test

Diode Drop Forward Bias _____
 Diode Drop Reverse Bias _____

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to *System Level Procedures: Troubleshooting: TLT Procedure*. Use Table DR-1 and DR-2 to record the TLT results of the defective coil.

TABLE DR-1
TLT DATA FOR DEFECTIVE SURFACE COIL

SITE: _____ NAME: _____ DATE: _____
 TLT FILE NUMBER: _____
 SNR: NOISE _____
 SNR MEAN _____
 SNR SIGNAL _____
 SNR AREA _____
 TR MAP: 89-91 _____ %
 85-95 _____ %
 65-115 _____ %
 FLIP MEAN _____

TABLE DR-2
TLT DATA FOR DEFECTIVE PHASED ARRAY COIL

SITE: _____ NAME: _____ DATE: _____

TLT FILE NUMBER: _____

SNR: NOISE (0) _____

NOISE (1) _____

NOISE (2) _____

NOISE (3) _____

NOISE (C) _____

SNR MEAN (0) _____

SNR MEAN (1) _____

SNR MEAN (2) _____

SNR MEAN (3) _____

SNR MEAN (C) _____

TR MAP: 89-91 (C) _____ %

(Combined 85-95 (C) _____ %

Results) 65-115 (C) _____ %

FLIP MEAN (C) _____

FLIP SDV (C) _____

RCV MEAN (C) _____

RCV SDV (C) _____

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Aug 19, 1998	R. Hawthorne	Initial conversion to Word
1	May 19, 1999	S.M.Atladottir	Updated Procedure Preferences for New GUI
2	Oct. 24, 2001	K. LaBarge	Updated procedure for SPT phantom and loader